Renal Dose Adjustment Protocol

Reference	PHA-REN-02 v4
Owner	Chief Pharmacist
Approved by	Quality and Patient Safety Committee
Effective date	02/03/2026
Next review	02/03/2029
Applies to	All clinical areas of the Centre Hospitalier Saint-Aubin

1. Purpose and scope

This protocol sets out how renal function is estimated and how the doses of renally cleared medicines are adjusted for adult inpatients. It applies to prescribers, ward pharmacists and nurses. Children, dialysis patients and patients with acute kidney injury follow separate guidance.

2. Estimating renal function

Renal function is expressed as the estimated glomerular filtration rate (eGFR) calculated with the CKD-EPI 2021 equation: **eGFR = 142 x min(Scr/kappa, 1)^alpha x max(Scr/kappa, 1)^-1.200 x 0.9938^Age x 1.012 [if female]**, in mL/min/1.73 m2.

Term	Rule
Serum creatinine in µmol/L	divide by 88.4 to obtain mg/dL (Scr)
Female	kappa = 0.7, alpha = -0.241, multiply the result by 1.012
Male	kappa = 0.9, alpha = -0.302
Age	whole years on the day of the sample

The eGFR is rounded to the nearest whole number before the dose band is read. The laboratory reports creatinine only; the eGFR is calculated by the prescriber or the ward pharmacist and written in the notes.

3. Enoxaparin, treatment dose

eGFR band (mL/min/1.73 m2)	Treatment dose	Monitoring
30 or above	1 mg/kg every 12 hours	none routinely
15 to 29	1 mg/kg once daily	anti-Xa level after the third dose
below 15	do not use; discuss with the pharmacist	-

Use the actual body weight in kilograms at admission. The calculated dose is rounded to the nearest prefilled syringe (20, 40, 60, 80, 100, 120 or 150 mg).

4. Other renally cleared medicines

Medicine	eGFR band	Adjustment
Metformin	45 or above	no change
Metformin	30 to 44	maximum 1,000 mg per day
Metformin	below 30	stop
Gentamicin	below 50	extended interval, level-guided
Apixaban (atrial fibrillation)	15 to 29	use with caution; 2.5 mg twice daily if two of age 80 or more, weight 60 kg or less, creatinine 133 micromol/L or more
Ramipril	below 30	start at 1.25 mg and titrate

5. Responsibilities

The prescriber records the eGFR, the band and the resulting dose in the notes. The ward pharmacist verifies the calculation at the next review and escalates unresolved discrepancies to the consultant.

6. Governance, audit and training

This document is controlled. Printed copies are uncontrolled: the current version is held on the hospital document management system and a printed copy must be checked against it before use.

The document owner reviews the content at least every three years or earlier when national guidance, a serious incident or an audit finding requires it. Interim changes are approved by the relevant committee chair and recorded in the revision history.

Compliance is audited by the clinical audit department on a rolling sample of records. Results are reported to the quality and patient safety committee, which agrees actions with the ward managers and monitors their completion.

Staff who identify a deviation from this document, or a situation it does not cover, should discuss it with the senior clinician on duty and report it through the incident reporting system. Reporting is a learning tool and not a disciplinary process.

7. Revision history

Version	Date	Author	Summary of change
1.0	02/03/2021	Chief Pharmacist	First issue
2.0	02/03/2023	Chief Pharmacist	Triennial review; format harmonised
3.0	02/06/2025	Chief Pharmacist	Tables updated after clinical audit
current	02/03/2026	Chief Pharmacist	Review and re-issue

Annex A. Implementation notes

This document is controlled. Printed copies are uncontrolled: the current version is held on the hospital document management system and a printed copy must be checked against it before use.

Equality and diversity: this document has been assessed for its effect on patients and staff with protected characteristics. Reasonable adjustments, interpreters and accessible formats are available through the patient experience team.

Training on this document is part of the induction programme for clinical staff and is refreshed through the annual mandatory training cycle. Agency and locum staff receive a short briefing at the start of their first shift.

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Data protection: any patient information handled under this document is processed in accordance with the data protection policy of the hospital. Access to records is limited to staff involved in the care of the patient.

Where this document refers to a role (for example the ward pharmacist or the senior nurse), the post-holder on duty carries out the task or delegates it to a suitably trained colleague who remains accountable to them.

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